CSE 21 Fall 2026
Homework 1
Submitted: October 12, 2026

Names: David Washington, Judah Boyce, Osinachi Agaranna, Nick Young
TSN: 200484943, 200453537, 200455712, 200455733

1. (a) Assume that for all $n \geq 1$

$$2 + 5 + 8 + \cdots + (3n - 1) = \frac{n(3n + 1)}{2}$$

First we prove the base case: $P(1)$ holds. When $n = 1$:

$$(3(1) - 1) = 2 = \frac{1(3(1) + 1)}{2} = \frac{4}{2} = 2$$

Assuming $P(k)$ is true. For $n \geq 1$, $P(k) \rightarrow P(k + 1)$ is again equivalent for the left and right side. RHS $P(k + 1)$

$$\frac{(k + 1)(3(k + 1) + 1)}{2} = \frac{(k + 1)(3k + 4)}{2} = \frac{3k^2 + 7k + 4}{2}$$

Since the LHS and RHS are equivalent we add the next element of the LHS to both sides and it should be equivalent to the RHS P(k+1)

$$2+5+8+\cdots+(3k-1)+(3(k+1)-1) = \frac{k(3k+1)}{2}+(3(k+1)-1)$$

$$= \frac{(k)(3k+1)}{2} + \frac{6k+4}{2}$$

$$= \frac{3k^2+k}{2} + \frac{6k+4}{2}$$

$$= \frac{3k^2+7k+4}{2}$$

The last equation shows that $P(k + 1)$ is true under the assumption that $P(k)$ is also true. Therefore, for all $n \geq 1$

$$2 + 5 + 8 + \cdots + (3n - 1) = \frac{n(3n + 1)}{2}$$

(b) Assume for all $n \geq 1$:

$$\prod_{k=1}^{n} \left(1 + \frac{1}{k}\right) = n + 1$$

First we prove that at $n = 1$ the left and right side are equivalent $P(1)$ holds.

$$\prod_{j=1}^{1} \left(1 + \frac{1}{j}\right) = 1 + 1 = 2$$

and the right side is $1 + 1 = 2$, so both sides match.

Assuming $P(k)$ is true. For $n \geq 1$, $P(k) \to P(k+1)$ is again equivalent for the left and right side. RHS $P(k+1)$

$$(k+1) + 1 = k + 2$$

Since we are assuming $P(k)$, we use the last factor of the product for the LHS and multiply it with the RHS and it should be equivalent to the RHS of $P(k+1)$:

$$\begin{aligned}
\prod_{j=1}^{k+1} \left(1 + \frac{1}{j}\right) &= \left[\prod_{j=1}^{k} \left(1 + \frac{1}{j}\right)\right] \left(1 + \frac{1}{k+1}\right) \\
&= (k+1)\left(1 + \frac{1}{k+1}\right) \\
&= (k+1) \cdot \frac{k+2}{k+1} \\
&= k + 2
\end{aligned}$$

This proves that $P(k+1)$ is true assuming that $P(k)$ is true. Therefore, by mathematical induction,

$$\prod_{k=1}^{n} \left(1 + \frac{1}{k}\right) = n + 1$$

for all $n \geq 1$.

2. A license plate is a string over the alphabet of 36 characters consisting of 26 uppercase Latin letters and 10 numerals (0 through 9). (2 points for a correct expression and 2 points for justification)

   (a) How many 7-character license plates have at least 2 numerals?

   The total number of combinations for 7 positions and 36 characters is 36^7. To find the case where there is at least 2 we can find the compliment (number of combos with no numbers or one number) and subtract that from the total.

   - Total number with no numbers: 26^7
   - Total number with one number: $\binom{7}{1} \cdot 10 \cdot 26^6$

   Number of license plates that have at least 2 numerals:

   $$= 36^7 - 26^7 - \binom{7}{1} \cdot 10 \cdot 26^6$$

(b) How many 7-character license plates contain the word MILES as a consecutive substring?

To find the possible combos, MILES is a 5 character string that must be in order in the license plate thus there are only 3 ways to place the specific string on the license plate. There are always 2 empty characters that allows for 36 characters each thus:

- Number of MILES placements: 3
- Number of unique options for 2 spots with 36 characters: 36^2

Number of plates that contain the word MILES as a consecutive substring:

$$= 3 \cdot 36^2$$

(c) How many 7-character license plates consist of 3 different characters with 3 copies of one, 3 copies of another and a single copy of the last? Ex: AAA3331, P33T3TT, ...

Choose 3 characters. Two of them appear 3 times, and the other appears once.

- Choose 3 characters from 36: $\binom{36}{3}$
- Choose which of those 3 appears once: 3
- Arrange 7 spots with those repetitions: $\frac{7!}{3! \cdot 3! \cdot 1!}$

Number of ways to arrange 3 different characters with 3 copies of one, 3 copies of another and a single copy of the last:

$$= \binom{36}{3} \cdot 3 \cdot \frac{7!}{3! \cdot 3! \cdot 1!}$$

(d) How many 7-character license plates have exactly 3 different numerals and 4 copies of the same letter?

Similar to the previous one you are choosing 1 character but in this case an alphabetical character from 26 which is repeated 4 times also in this problem you have the last 3 characters as unique numerals

- Choose the repeated letter: $\binom{26}{1}$
- Choose 3 different numerals: $\binom{10}{3}$
- Arrange 7 spots with that letter repeated 4 times: $\frac{7!}{4! \cdot 1!}$

The number of ways to have exactly 3 different numerals and 4 copies of the same letter:

$$= \binom{26}{1} \cdot \binom{10}{3} \cdot \frac{7!}{4! \cdot 1!}$$

(e) How many 7-character license plates are there that are all letters and they are in alphabetical order?

For this one you choose 7 alphabetical characters from 26 and that set has one alphabetical order.

- Choose the 7 from 26: $\binom{26}{7}$

The number of ways that the license plate is all letters and they are in alphabetical order:

$$= \binom{26}{7}$$

3. An art museum with 12 different rooms just got a shipment of 9 new paintings

   (a) How many ways can the museum curator arrange all 9 paintings among the 12 rooms in the museum? You can arrange 9 paintings and have each painting have 12 options to choose form when being placed. Thus the number of ways to place 9 paintings amon 12 rooms in the museum is:

   $$= 12^9$$

   (b) How many ways can the museum curator arrange all 9 paintings among the 12 rooms in the museum so that each painting is in a different room? This question asks the number of permutations of 9 paintings in 12 rooms.

   Thus the number of ways to place 9 paintings among 12 rooms in the museum is:

   $$= P(12, 9)$$

   (c) How many ways can the museum curator arrange all 9 paintings among the 12 rooms in the museum so that at least one painting is in the biggest room and exactly one painting is in the smallest room? For this question we consider the one painting chosen for the the smallest room.

   - Number of ways to chosen 1 painting for the smallest room = $\binom{9}{1}$
   - Number of ways to place 8 paintings in any of the other 11 rooms (compliment, total arrangements subtracted by possible arrangements not in the biggest room) = $11^8 - 10^8$

   So the total number of arrangements for 9 paintings one in the biggest room and at least one in the smallest is:

   $$= \binom{9}{1} \cdot (11^8 - 10^8)$$

   (d) Paintings A,B, and C are all painted by the same artist. How many ways can the museum curator arrange all 9 paintings among the 12 rooms such that paintings A,B, and C are all in the same room? We can consider paintings A, B, and C to be one set or basically one painting and so now we have 7 paintings being placed in 12 rooms So the total number of arrangements for all 9 paintings when A,B, and C must be in the same room is:

   $$= 12^7$$

(e) How many ways can the museum curator arrange all 9 paintings in 3 of the 12 rooms such that each room has exactly 3 paintings? Given that each room has exactly 3 paintings it means that 3 rooms will be used at a time, so we can think of it in the perspective of those 3 rooms.

- First you have 12 rooms and you need to choose 3 rooms = $\binom{12}{3}$
- For the first room you have 9 paintings now and need 3 = $\binom{9}{3}$
- Same thing for the second = $\binom{6}{3}$
- Same thing for the third = $\binom{3}{3}$

Thus the total number of ways is equal to product of each room:

$$= \binom{12}{3} \cdot \binom{9}{3} \cdot \binom{6}{3} \cdot \binom{3}{3}$$

4. (a) How many different ways are there to arrange the letters in the following string:

INCLUSIONEXCLUSION

The string contains 18 letters total, with a repeat of letters with the following multiplicities:

$$I : 3, \quad N : 3, \quad C : 2, \quad L : 2, \quad U : 2, \quad S : 2, \quad O : 2.$$

The letters E and X each occur once. Since rearranging identical copies of the same letter not produce a new arrangement we must divide by the factorial of the repeats of each letter:

$$\boxed{\frac{18!}{3!3!(2!)^5}}$$

Justification: We divide by 3! twice for the three copies of I and N, and by 2! for each of the five letters that appear twice. This removes all duplicate arrangments.

(b) How many different ways are there to color the 9 edges of a triangular prism with 9 different colors, where two colorings are considered the same if one can be oriented to look the same as the other?

If we first look at the amount of ways we could assign the 9 distinct colors to be assigned to the 9 distict edges we would get

$$9!$$

ways.

A triangular prism has 6 rotational symmetries and since there are 9 different colors no non-identiy rotation can leave a coloring unchanged. As a result, each distinct coloring is counted 6 times among the arrangements:

$$\boxed{\frac{9!}{6}}$$

Justification: The 6 possible rotations of the triangular prism produce colorings that are considered equivalent and because every edge has a different color, each group contains 6 rotated versions.

(c) There are 18 different basketball players. How many ways can they organize themselves into 3 games, each with 3 players versus 3 players?

$$\boxed{\frac{18!}{(3!)^6 2^3 3!}}$$

Justification: The factor $(3!)^6$ accounts for the fact that the order of the 3 players within each team does not matter. The factor 2^3 accounts for swapping the two teams in each game. 3! accounts for rearranging the order of the three games .